

Module 5

Valuing Bonds and Stocks

FINE 3010, Financial Management
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From Module 4 to Module 5

- **Module 4**, the time value of money, the present value of future cash flows

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- **Module 4**, the time value of money, the present value of future cash flows
- **Module 5**, the same time-value-of-money principles value bonds and stocks
 - Bond, the present value of its coupons and face value
 - Stock, the present value of its expected dividends

Every Financial Asset Is Debt or Equity

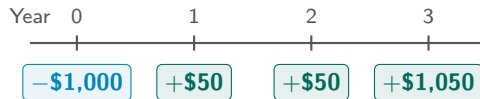
- **Financial asset**, value from a contractual right or ownership claim



Debt's Cash Flow Is Fixed, a Stock's Is Not

Debt

\$1,000, 3 years, 5% interest

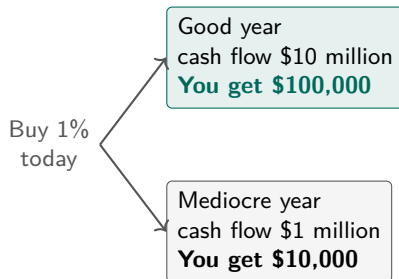


Every payment known in advance

Stock

1% of the company

Next year



Value of claim depends on how the company does

Common Types of Corporate Financing

Debt, lending to the company

Main forms, bonds and loans

Holder gets a fixed promised amount

“Senior”, paid back before equity in
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No voting rights or control

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Equity, stock ownership in the company

Holder gets dividends

“Residual claimant”, paid whatever is left after all promised payments, no guaranteed payout

Voting rights

Module 5, Part 1

Bonds

Bonds, Definitions

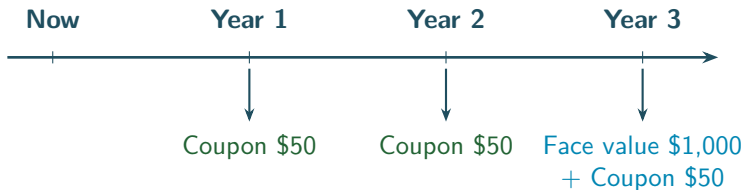
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Bonds, Definitions

- **Bond**, a security that obligates the issuer (the borrower) to make specified payments to the investor (the lender, or “bondholder”)
- Typically some periodic payments (the “coupon”), then a “face value” at the end
- E.g., a three-year bond with an annual coupon of \$50 and a face value of \$1,000, the borrower promises to pay,



Bonds, Terminology

- **Face value**, the final lump-sum payment at the “maturity” of the bond, usually \$1,000 per bond in the U.S.
 - **Not** the same as the price of the bond

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- **Coupon rate**,

$$\text{Coupon rate} = \frac{\text{Annual coupon}}{\text{Face value}}$$

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 - E.g., **Module 4**'s Puerto Rico bond, the coupon was 0 but the interest rate was 5.15%

Example, Puerto Rico Borrows

Recall from **Module 4**

- In 2007, Puerto Rico sold bonds that promised to pay \$1,000 after 47 years (no periodic “coupon” payments)
- These bonds were sold at \$94.40 each
- What was the interest rate paid by Puerto Rico?



Example, Puerto Rico Borrows

Recall from **Module 4**



$$N = 47, \quad I = ?$$

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- Solution,
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 - -94.40 **PV**
 - 0 **PMT**
 - 1000 **FV**
 - **I/Y**

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- The rate is 5.15%

The Equation the Calculator Solves

Recall from **Module 4**

- Recall, your financial calculator is programmed to solve this equation

$$PV + \left[\frac{PMT}{(1+I)^1} + \frac{PMT}{(1+I)^2} \cdots + \frac{PMT}{(1+I)^N} \right] + \frac{FV}{(1+I)^N} = 0$$

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The present values of PV, all PMTs, and FV, summed up together equal 0.

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$$\text{Value} = \text{PV} = \frac{\text{Coupon}_1}{(1+r)^1} + \frac{\text{Coupon}_2}{(1+r)^2} + \dots + \frac{\text{Coupon}_T + \text{Face value}}{(1+r)^T}$$

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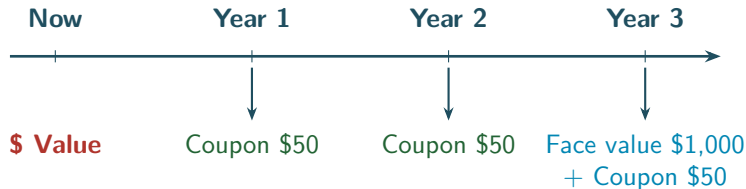
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- Investors choose the required rate of return r as the opportunity cost, “what could I get investing in something else with a similar risk”

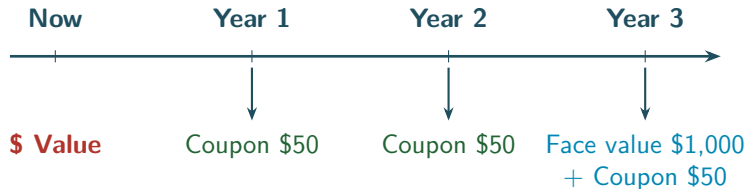
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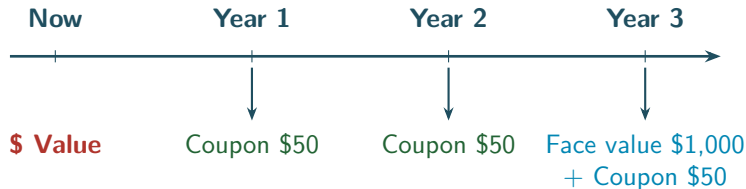
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- What is the value of the bond if the required rate of return r is 4%?

$$\text{Value} = \frac{\$50}{(1 + 0.04)^1} + \frac{\$50}{(1 + 0.04)^2} + \frac{\$50 + \$1,000}{(1 + 0.04)^3} = \$1,027.75$$

Example, Bond Pricing with a Financial Calculator

- The same bond, an annual coupon of \$50, a face value of \$1,000, three years to maturity, and a required rate of return of 4%
- Use a financial calculator to value the bond

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 - 1000 **FV**
 - 50 **PMT**
 - 3 **N**
 - 4 **I/Y**
 - **CPT**, then **PV**

Example, Bond Pricing with a Financial Calculator

- The same bond, an annual coupon of \$50, a face value of \$1,000, three years to maturity, and a required rate of return of 4%
- Use a financial calculator to value the bond
- Solution,
 - 1000 **FV**
 - 50 **PMT**
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 - **CPT**, then **PV**
- The output is $-\$1,027.75$ (why the minus sign? recall how financial calculators get to the answer in [Module 4](#))

Valuing Bonds with a Financial Calculator

- The financial calculator keys map onto a bond,
 - **N**, the number of periods (years) until maturity
 - **PV**, the value of the bond (the present value of the coupons and the face value)
 - **PMT**, the coupons (in dollars)
 - **FV**, the face value (the lump sum paid at the end)
 - **I/Y**, the discount rate (the required rate of return r)

Recap, Bond Pricing

- Companies issue bonds (and other debt) to raise financing from investors

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- In return, the companies make pre-specified payments of coupons and face value to the investors

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- Companies issue bonds (and other debt) to raise financing from investors
- In return, the companies make pre-specified payments of coupons and face value to the investors
- The value of a bond is the discounted payments, at the required rate of return

Bond Yields

- **Yield to maturity (YTM)**, the discount rate y for which the present value of the bond's payments (coupons and face value) equals its price (also simply the “yield”)

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- **Yield to maturity (YTM)**, the discount rate y for which the present value of the bond's payments (coupons and face value) equals its price (also simply the "yield")
- Given the price of a bond (PV), its maturity T , and its coupons, find the yield to maturity by solving for y ,

$$\text{Price} = \frac{\text{Coupon}}{(1+y)^1} + \frac{\text{Coupon}}{(1+y)^2} + \dots + \frac{\text{Coupon} + \text{Face value}}{(1+y)^T}$$

Example, Bond Yields

- Solving for the YTM analytically is hard, so use the financial calculator
- A bond's annual coupon rate is 5%. Its face value is \$1,000. Its maturity is 3 years. Its current market price is \$1,027.75. What is its YTM?

Example, Bond Yields

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- A bond's annual coupon rate is 5%. Its face value is \$1,000. Its maturity is 3 years. Its current market price is \$1,027.75. What is its YTM?
- Solution,
 - 1000 **FV**
 - 50 **PMT**
 - 3 **N**
 - -1027.75 **PV**
 - **CPT**, then **I/Y**

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- A bond's annual coupon rate is 5%. Its face value is \$1,000. Its maturity is 3 years. Its current market price is \$1,027.75. What is its YTM?
- Solution,
 - 1000 **FV**
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 - **CPT**, then **I/Y**
- The yield is 4%

Practice Problem

- The table shows data for three bonds. Each bond's coupon is zero. Each bond's face value is \$1,000

Bond	Price	Maturity (years)	Yield to maturity
A	\$310	25	?
B	\$310	?	9%
C	?	11	8%

- What is the yield to maturity of bond A?
- What is the maturity of bond B?
- What is the price of bond C?

Practice Problem

- Solution, on the financial calculator (the highlighted key is the one solved for),

	N	I/Y	PV	PMT	FV
a. Bond A	25	?	-310	0	1,000
b. Bond B	?	9	-310	0	1,000
c. Bond C	11	8	?	0	-1,000

Practice Problem

- Solution, on the financial calculator (the highlighted key is the one solved for),

	N	I/Y	PV	PMT	FV
a. Bond A	25	4.796	-310	0	1,000
b. Bond B	?	9	-310	0	1,000
c. Bond C	11	8	?	0	-1,000

- a. The yield to maturity of bond A is 4.796%

Practice Problem

- Solution, on the financial calculator (the highlighted key is the one solved for),

	N	I/Y	PV	PMT	FV
a. Bond A	25	4.796	-310	0	1,000
b. Bond B	13.59	9	-310	0	1,000
c. Bond C	11	8	?	0	-1,000

- a. The yield to maturity of bond A is 4.796%
- b. The maturity of bond B is 13.59 years

Practice Problem

- Solution, on the financial calculator (the highlighted key is the one solved for),

	N	I/Y	PV	PMT	FV
a. Bond A	25	4.796	-310	0	1,000
b. Bond B	13.59	9	-310	0	1,000
c. Bond C	11	8	428.88	0	-1,000

- a. The yield to maturity of bond A is 4.796%
- b. The maturity of bond B is 13.59 years
- c. The price of bond C is \$428.88

Practice Problem

- A firm's bonds pay an annual coupon rate of 6.8%. Their face value is \$1,000. Their price is \$795.69. Their yield to maturity is 9.50%
- What is the remaining maturity of these bonds?

Practice Problem

- Solution, the bond price is the PV of the coupon payments plus the PV of the face value,

Bond price = PV of coupon payments + PV of face value

Bond price = $C \times ((1 \div r) - \{1 \div [r(1 + r)^t]\}) + FV \div (1 + r)^t$

$\$795.69 = (0.068 \times \$1,000) \times ((1 \div 0.095) - \{1 \div [0.095 \times (1 + 0.095)^t]\}) + \$1,000 \div (1.095)^t$

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- On the financial calculator,
 - 9.5 **I/Y**, -795.69 **PV**, 68 **PMT**, 1000 **FV**, then **CPT** **N**

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- On the financial calculator,
 - 9.5 **I/Y**, -795.69 **PV**, 68 **PMT**, 1000 **FV**, then **CPT** **N**
- The remaining maturity is about 14 years ($N = 13.98$)

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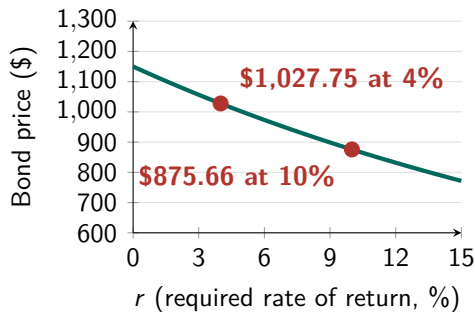
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- What determines r (the required rate of return, or expected return)?
 1. Prevailing (risk-free) interest rates (the opportunity cost for time value)
 - Higher prevailing (risk-free) interest rate, higher r
 2. Risk of default (the opportunity cost for risk-taking)
 - Higher (systematic) risk of default, higher r

Interest Rates and Bond Prices Move in Opposite Directions

- Because the bond price is,

$$\text{Price} = \frac{\text{Coupon}_1}{(1+r)^1} + \dots + \frac{\text{Coupon}_T + \text{Face value}}{(1+r)^T}$$

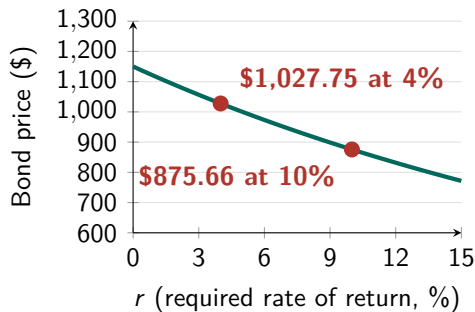


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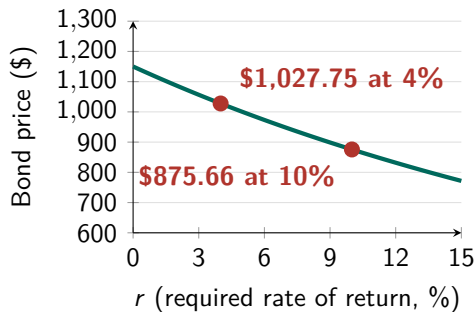


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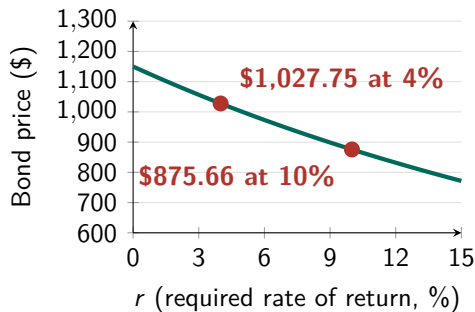


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- **Default risk (credit risk)**, the risk that a bond issuer may default on its bonds

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- **Default premium**, the additional yield on a bond that investors require for bearing credit risk

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- **High-yield bonds** ("junk" bonds), rated Ba or BB and lower

Default Risk, Credit Ratings

Moody's	S&P	Default within 10 yr
Investment-grade bonds		
Aaa	AAA	0.1%
Aa	AA	0.7%
A	A	2.2%
Baa	BBB	3.5%
High-yield bonds		
Ba	BB	15.7%
B	B	35.5%
Caa, Ca	CCC, CC	48.9%
C	C	n/a

Source, Brealey, Myers, and Marcus, Table 6.2

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- Yields tend to be higher (prices lower) for lower-rated bonds

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- Yield to maturity, the discount rate for which the present value of the bond's payments equals its price
- Interest rates and bond prices move in opposite directions
- r rises with prevailing interest rates and with default risk, lower-rated bonds carry higher yields

Module 5, Part 2

Stocks

Goals

- **Part 1, bonds**
 - Differences between debt and equity
 - Valuation
 - Yields
 - Interest rates
 - Default risk
- **Part 2, stocks**
 - Valuation, dividend discount model
 - Valuation, comparables
 - Market efficiency

Valuing Stocks

- Two alternative ways to value stocks,
 1. Dividend discount model
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 1. Dividend discount model
 2. Valuation by comparables
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- You will learn more in “FINE 4100 Advanced Financial Management” and “FINE 4110 Investments in Equities”

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Stock Valuation, Terminology

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- **Dividend yield**, $\frac{\text{Dividend}}{\text{Stock price}}$
- **Earnings per share (EPS)**, $\frac{\text{Net income}}{\text{Number of shares outstanding}}$
- **P/E ratio**, $\frac{\text{Stock price}}{\text{Earnings per share}}$

Valuing Stocks, Dividend Discount Model

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- What price should you pay for the stock today (P_0)?

Valuing Stocks, Dividend Discount Model

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- What price should you pay for the stock today (P_0)?
- Stock value, the present value of the cash flows you get from owning it,

$$P_0 = \frac{\text{Div}_1}{(1 + r_e)^1} + \frac{\text{Div}_2}{(1 + r_e)^2} + \frac{\text{Div}_3}{(1 + r_e)^3} + \cdots + \frac{\text{Div}_\infty}{(1 + r_e)^\infty} = \text{PV}(\text{Dividends})$$

Valuing Stocks, Dividend Discount Model

- Suppose you plan to buy a stock and own it forever
- What price should you pay for the stock today (P_0)?
- Stock value, the present value of the cash flows you get from owning it,

$$P_0 = \frac{\text{Div}_1}{(1 + r_e)^1} + \frac{\text{Div}_2}{(1 + r_e)^2} + \frac{\text{Div}_3}{(1 + r_e)^3} + \cdots + \frac{\text{Div}_\infty}{(1 + r_e)^\infty} = \text{PV}(\text{Dividends})$$

- **Required return on equity** (r_e), the discount rate, set by the opportunity cost of investing in something with similar risk

Valuing Stocks, Dividend Discount Model

- Suppose you plan to buy a stock and own it forever
- What price should you pay for the stock today (P_0)?
- Stock value, the present value of the cash flows you get from owning it,

$$P_0 = \frac{\text{Div}_1}{(1 + r_e)^1} + \frac{\text{Div}_2}{(1 + r_e)^2} + \frac{\text{Div}_3}{(1 + r_e)^3} + \cdots + \frac{\text{Div}_\infty}{(1 + r_e)^\infty} = \text{PV}(\text{Dividends})$$

- **Required return on equity** (r_e), the discount rate, set by the opportunity cost of investing in something with similar risk
- $\text{Div}_1, \text{Div}_2, \dots$ are *expected* dividends

Gordon Growth Model

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- The latter formula is often called the “**Gordon growth model**”

Example, Gordon Growth Model

- 1. A firm will pay a \$3.00 dividend forever (the first dividend exactly a year from now). Its required return on equity is 12%. What will investors pay for the stock?
- 2. What if the firm's stock pays a \$3.00 dividend a year from now, and then the dividend grows at 8% forever?

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- Stock value today, the present value of these cash flows,

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- That is, dividend yield plus expected capital appreciation (in %)

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- Uncertain dividend forecasts
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- How to value non-dividend-paying stocks?
 - If the firm is only temporarily not paying dividends, the dividend discount model still applies
 - But future dividends are then probably even more uncertain!

Valuing Dividends with Non-Constant Growth

- Estimate the dividends year by year over a horizon of H years, then assume constant growth after year H ,

$$PV = \frac{\text{Div}_1}{(1 + r_e)^1} + \frac{\text{Div}_2}{(1 + r_e)^2} + \dots + \frac{\text{Div}_H}{(1 + r_e)^H} + \frac{P_H}{(1 + r_e)^H}$$

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- You will learn more in “FINE 4100 Advanced Financial Management” and “FINE 4110 Investments in Equities”

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- A stock's expected dividends are in the table. Its required return on equity is 8.5%. Its dividends grow at 6% a year beginning in year 6. What is the value of the stock?

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$$PV_{\text{Stock}} = 10.23 + 89.11 = \$99.33$$

Practice Problem

- Consider the following three stocks,
 - a. Stock A is expected to pay a dividend of \$10 a share forever
 - b. Stock B is expected to pay a dividend of \$5 next year. Thereafter, dividend growth is expected to be 4% a year forever
 - c. Stock C is expected to pay a dividend of \$5 next year. Thereafter, dividend growth is expected to be 20% a year for 5 years (years 2 through 6) and zero thereafter
- Each stock's required return on equity is 10%. Which stock is the most valuable? What if the required return on equity is 7%?

Practice Problem

- Solution, at a 10% required return on equity,

10% capitalization rate:

$$P_A = \text{DIV}_1 / r = \$10 / 0.1 = \$100$$

$$P_B = \text{DIV}_1 / (r - g) = \$5 / (0.1 - 0.04) = \$83.33$$

$$\begin{aligned} P_C &= \frac{\text{DIV}_1}{1+r} + \frac{\text{DIV}_1 \times (1+g)}{(1+r)^2} + \frac{\text{DIV}_1 \times (1+g)^2}{(1+r)^3} + \frac{\text{DIV}_1 \times (1+g)^3}{(1+r)^4} + \frac{\text{DIV}_1 \times (1+g)^4}{(1+r)^5} + \\ &\quad \frac{\text{DIV}_1 \times (1+g)^5}{(1+r)^6} + \frac{(\text{DIV}_1 \times [1+g]^5 \times [1+g_2]) / r}{(1+r)^6} \\ P_C &= \$5 / 1.1 + (\$5 \times 1.2) / 1.1^2 + (\$5 \times 1.2^2) / 1.1^3 + (\$5 \times 1.2^3) / 1.1^4 \\ &\quad + (\$5 \times 1.2^4) / 1.1^5 + (\$5 \times 1.2^5) / 1.1^6 + ([\$5 \times 1.2^5 \times \{1 + 0\}] / 0.1) / 1.1^6 \\ P_C &= \$104.51 \end{aligned}$$

At a 10% capitalization rate, Stock C has the largest present value.

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At a 10% capitalization rate, Stock C has the largest present value.

- At a 7% required return on equity, $P_A = \$142.86$, $P_B = \$166.67$, and $P_C = \$156.50$, so Stock B is the most valuable

Valuing Stocks, Comparables

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- E.g., a 2,000 sqft house in New Orleans sells for \$400,000. What is your estimate for a neighboring house of 2,500 sqft?

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- E.g., Company A creates twice the cash flow per share of Company B, and Company B trades at \$10. What price should Company A's shares trade at?
- Underlying idea, we can “scale” prices!

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- 4. Back out Company X’s implied value (the numerator) as the average multiple times Company X’s denominator

$$P_X = (\text{Average P/E of comparables}) \times \text{EPS}_X$$

Example, Valuation by Comparables

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- Solution, multiply Herman Miller's EPS by the P/E of comparable firms,
$$P = (\text{Average P/E of comparables}) \times \text{EPS} = 21.3 \times 1.38 = \$29.39$$

Example, Valuation by Comparables

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- The question asks for the share price, not the market cap,

$$P = \frac{\text{Market cap}}{\text{Number of shares outstanding}} = \frac{36.8 \text{ billion}}{1.2 \text{ billion}} = \$30.67$$

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 - Matters if the differences affect future cash flow growth or the risks the firms face
- Assumes the comparables are priced correctly
 - Cannot tell if an entire industry is overvalued or undervalued
 - E.g., no help in valuing tech firms during the late-1990s “dot-com bubble”

Buying and Selling Stock on the Stock Market

- Current stock price

FedEx Corporation (FDX)

☆ Add to watchlist

NYSE - Nasdaq Real Time Price. Currency in USD

239.98 -9.04 (-3.63%)

As of 1:02PM EDT. Market open.

Summary

Chart

Conversations

Statistics

Profile

Final

Previous Close	249.02	Market Cap	64.318B
Open	246.00	Beta	1.57
Bid	239.98 x 100	PE Ratio (TTM)	22.11
Ask	240.10 x 100	EPS (TTM)	10.86
Day's Range	236.00 - 246.00	Earnings Date	Mar 19, 2018 - Mar 23, 2018
52 Week Range	182.89 - 274.66	Forward Dividend & Yield	2.00 (0.80%)
Volume	2,359,358	Ex-Dividend Date	N/A
Avg. Volume	1,866,076	1y Target Est	284.69

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- **Bid**, the price at which investors are willing to buy shares
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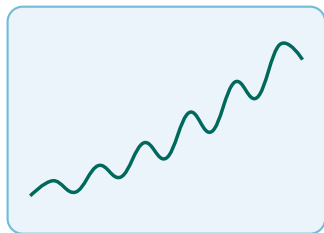
- Market capitalization
- **Beta**, a measure of risk (how much the stock moves with the overall market)
- Price/earnings
- EPS
- Dividend and dividend yield

Can You Beat the Stock Market?

- It's really hard!

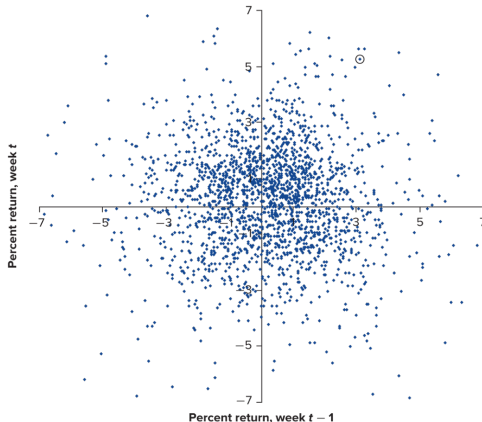
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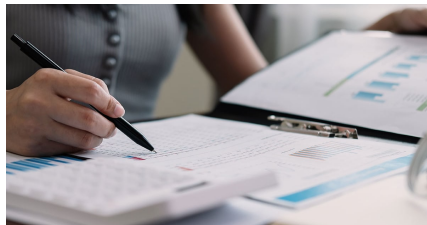
Discuss, Where Is the Pattern?

- Scatter plot of the NYSE Composite Index return in one week against its return in the previous week



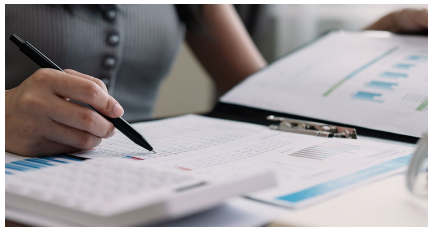
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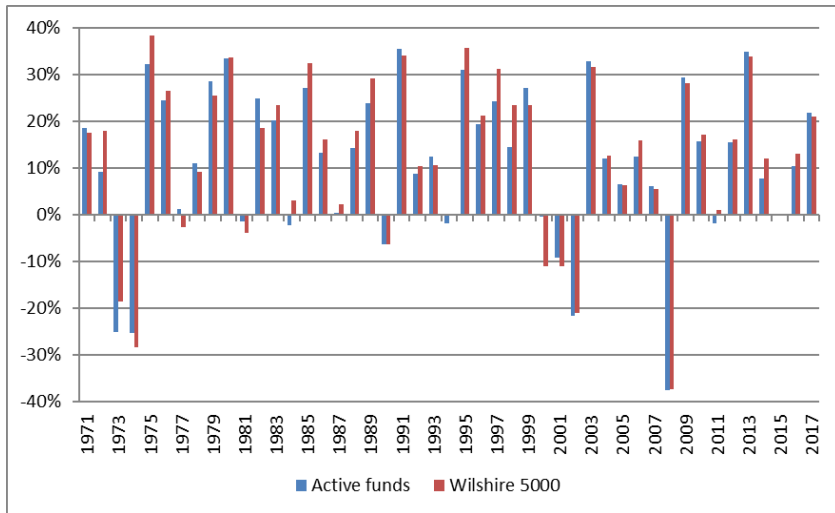


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Professional Fund Managers Don't Generally Beat the Index!



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Information in Stock Prices

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- **Efficient markets hypothesis**, “securities are fairly priced, given all information that is available to investors”
- What **type** of information is captured in the stock price?
 1. All historical price information? (“weak” efficiency)
 2. All public, easily available information? (“semi-strong” efficiency)
 3. Also all private or difficult-to-interpret information? (“strong” efficiency)

What Does This Mean for Investors?

Pros 👍

- If stocks are fairly priced, even an investor who **knows nothing** can buy **any** stock and expect it to fairly compensate them for the risk they take

Cons 👎

- To *consistently* make money above the *fair (risk-compensating)* return by betting on publicly traded shares, an investor needs a very distinct competitive advantage
- E.g., expertise or access to information that only a few people can figure out

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- Comparables, scale a peer multiple (P/E, market-to-book) to the firm being valued
- Beating the market is really hard, prices already reflect available information (the efficient markets hypothesis)

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- Interest rates and bond prices move in opposite directions
- r rises with prevailing interest rates and with default risk, lower-rated bonds carry higher yields
- Stocks, the dividend discount model or comparables, and efficient markets make the index hard to beat

Thank you!

Please let me know if you have any questions.